Enterprise Network

Cisco Packet Tracer - Step-by-Step Build Guide (Project 4)

Goal: build a two-building enterprise network: HR, Sales and Finance floors in each building, redundant gateways (HSRP), dynamic routing (OSPF) between the buildings over a serial link, and one central DHCP, DNS and web server. The server sits on its own dual-homed access switch (Access-SW7), so it stays connected if one multilayer switch fails.

You will use: 2 Routers, 4 Multilayer Switches, 7 Access Switches (6 floor switches + 1 server switch), 36 PCs (6 per floor), 6 Printers (1 per floor), 1 Server.

Two parts: Part 1 builds the network (VLANs, HSRP, OSPF). Part 2 adds the server VLAN, the server switch (Access-SW7) and the central DHCP, DNS and web server.

1. What you are building

Area	Devices	VLAN
Building 1 - HR floor	6 PCs + Printer 1 on Access-SW1	10
Building 1 - Sales floor	6 PCs + Printer 2 on Access-SW2	20
Building 1 - Finance floor	6 PCs + Printer 3 on Access-SW3	30
Building 2 - HR floor	6 PCs + Printer 1 on Access-SW4	40
Building 2 - Sales floor	6 PCs + Printer 2 on Access-SW5	50
Building 2 - Finance floor	6 PCs + Printer 3 on Access-SW6	60
Server room (Building 1)	Server: DHCP + DNS + web (Part 2), on Access-SW7	100 (Servers)
Network cabinet (Building 1)	1 Router + 2 Multilayer Switches + 4 Access Switches (3 floor switches + Access-SW7)	-
Network cabinet (Building 2)	1 Router + 2 Multilayer Switches + 3 Access Switches	-

Note: One Access Switch per floor, and one VLAN per floor. Each Access Switch has two uplinks, one to each multilayer switch. The server has its own Access Switch (Access-SW7) with the same two uplinks, so no single switch failure cuts the server off.

Key devices

Device	Hostname	Role
Router	R1 (Bldg 1), R2 (Bldg 2)	Connects the building to the other building (serial link) and runs OSPF
Multilayer Switches	L3-SW1, L3-SW2 (Bldg 1); L3-SW3, L3-SW4 (Bldg 2)	Gateways for the VLANs (HSRP pair), inter-VLAN routing, OSPF, DHCP relay
Access Switches	Access-SW1 - 3 (Bldg 1); Access-SW4 - 6 (Bldg 2)	Connect the floor's PCs and printer
Server Access Switch	Access-SW7 (Bldg 1)	Connects the server to both L3-SW1 and L3-SW2 (dual-homed), VLAN 100
Server	Server	Central DHCP, DNS and web server (Building 1, VLAN 100, on Access-SW7)

Note: HSRP gives each VLAN one virtual gateway shared by two switches. If the active switch fails, the standby switch takes over and the PCs keep the same default gateway.

2. Network plan (cheat sheet)

VLAN	Name	Subnet	Gateway (HSRP)	SVIs (.2 / .3)	Access switch
10	HR (Bldg 1)	192.168.10.0/24	192.168.10.1	L3-SW1 .2 / L3-SW2 .3	Access-SW1
20	Sales (Bldg 1)	192.168.20.0/24	192.168.20.1	L3-SW1 .2 / L3-SW2 .3	Access-SW2
30	Finance (Bldg 1)	192.168.30.0/24	192.168.30.1	L3-SW1 .2 / L3-SW2 .3	Access-SW3
40	HR (Bldg 2)	192.168.40.0/24	192.168.40.1	L3-SW3 .2 / L3-SW4 .3	Access-SW4
50	Sales (Bldg 2)	192.168.50.0/24	192.168.50.1	L3-SW3 .2 / L3-SW4 .3	Access-SW5
60	Finance (Bldg 2)	192.168.60.0/24	192.168.60.1	L3-SW3 .2 / L3-SW4 .3	Access-SW6
100	Servers (Bldg 1)	192.168.100.0/24	192.168.100.1	L3-SW1 .2 / L3-SW2 .3	Server on Access-SW7 F0/3
100	Servers (Bldg 2)	192.168.110.0/24	192.168.110.1	L3-SW3 .2 / L3-SW4 .3	no server yet

VLAN 100 is used in both buildings, but they are separate networks with different subnets. Printers use static addresses (.20); PCs use DHCP from the server (.100 and up).

HSRP priorities

Switch	VLANs 10-30 / 40-60	VLAN 100	Role
L3-SW1	110	210	HSRP active
L3-SW2	100	200	HSRP standby
L3-SW3	210	210	HSRP active
L3-SW4	200	200	HSRP standby

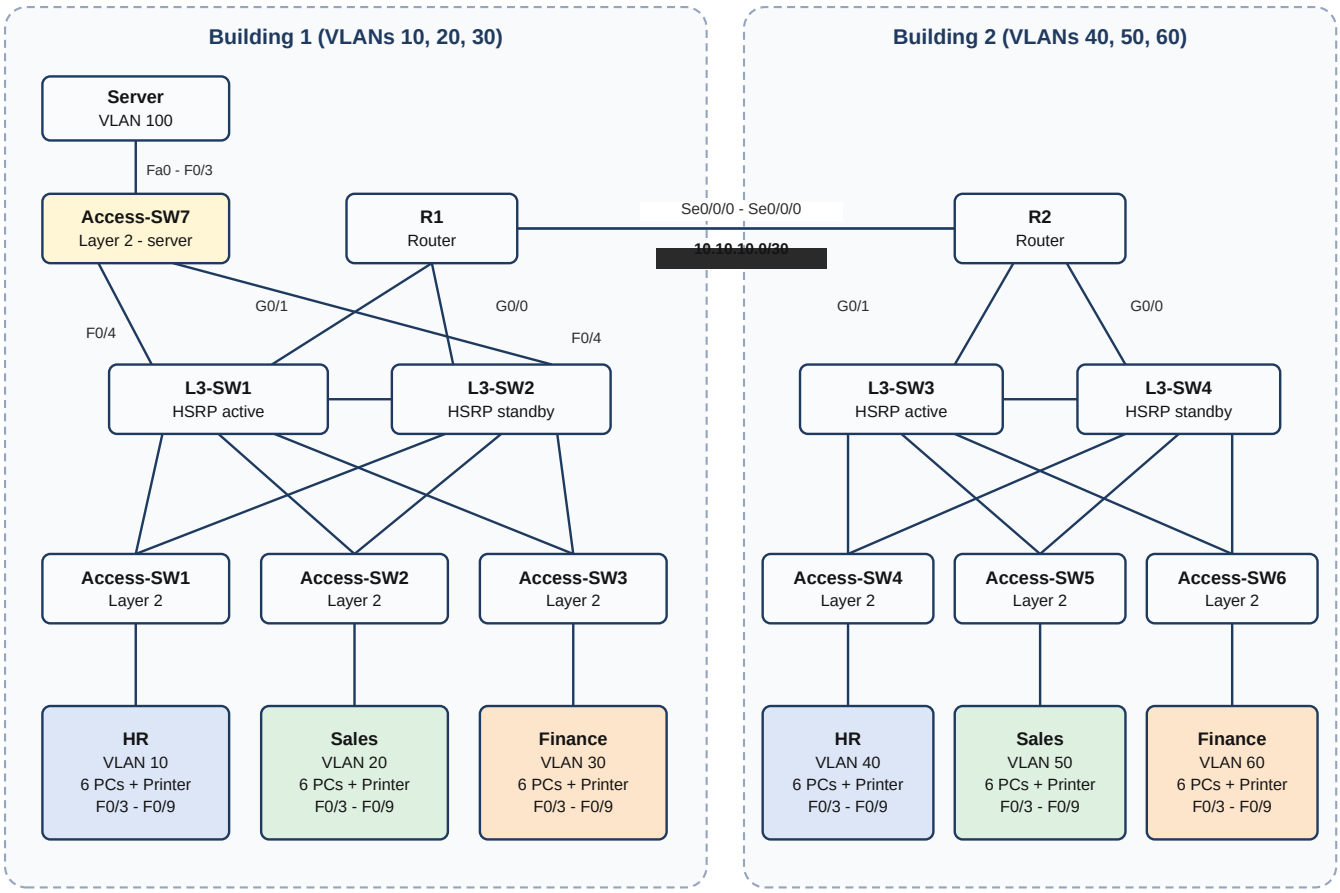
Note: The higher priority wins. 'preempt' lets a switch take the active role back after it comes back up.

Routed links (OSPF process 1, area 0)

Link	Subnet	Addresses
R1 G0/1 - L3-SW1 G0/1	10.10.1.0/30	R1 = 10.10.1.1; L3-SW1 = 10.10.1.2
R1 G0/0 - L3-SW2 G0/1	10.10.1.4/30	R1 = 10.10.1.5; L3-SW2 = 10.10.1.6
R2 G0/1 - L3-SW3 G0/1	10.10.2.0/30	R2 = 10.10.2.1; L3-SW3 = 10.10.2.2
R2 G0/0 - L3-SW4 G0/1	10.10.2.4/30	R2 = 10.10.2.5; L3-SW4 = 10.10.2.6
R1 Se0/0/0 - R2 Se0/0/0	10.10.10.0/30	R1 = 10.10.10.1 (DCE, clock rate 64000); R2 = 10.10.10.2

Note: Server (Building 1): 192.168.100.10 /24, gateway 192.168.100.1, DNS = itself (192.168.100.10).

Topology



Access-SW7 (yellow) is the new server switch: it connects the server to both L3-SW1 and L3-SW2 (F0/4 on each).

3. Physical connections

Routers and multilayer switches

#	From	To	Cable
1	R1 Se0/0/0	R2 Se0/0/0	Serial DCE (DCE end on R1)
2	R1 G0/1	L3-SW1 G0/1	Ethernet
3	R1 G0/0	L3-SW2 G0/1	Ethernet
4	L3-SW1 G0/2	L3-SW2 G0/2	Ethernet
5	R2 G0/1	L3-SW3 G0/1	Ethernet
6	R2 G0/0	L3-SW4 G0/1	Ethernet
7	L3-SW3 G0/2	L3-SW4 G0/2	Ethernet

Server switch (Access-SW7, Building 1)

#	From	To	Cable
8	Access-SW7 F0/1	L3-SW1 F0/4	Ethernet
9	Access-SW7 F0/2	L3-SW2 F0/4	Ethernet
10	Server Fa0	Access-SW7 F0/3	Ethernet

Note: The server is not connected to L3-SW1 F0/24 any more. Access-SW7 has two uplinks, one to each multilayer switch, like the floor switches. If one multilayer switch fails, the server stays connected through the other one.

Access switches (uplinks, PCs and printers)

Access switch	Floor (VLAN)	F0/1 to	F0/2 to	F0/3 - F0/8	F0/9
Access-SW1	HR (10)	L3-SW1 F0/1	L3-SW2 F0/1	PC 1 - 6	Printer 1
Access-SW2	Sales (20)	L3-SW1 F0/2	L3-SW2 F0/2	PC 7 - 12	Printer 2
Access-SW3	Finance (30)	L3-SW1 F0/3	L3-SW2 F0/3	PC 13 - 18	Printer 3
Access-SW4	HR (40)	L3-SW3 F0/1	L3-SW4 F0/1	PC 1 - 6	Printer 1
Access-SW5	Sales (50)	L3-SW3 F0/2	L3-SW4 F0/2	PC 7 - 12	Printer 2
Access-SW6	Finance (60)	L3-SW3 F0/3	L3-SW4 F0/3	PC 13 - 18	Printer 3

Note: PC 1 = F0/3, PC 2 = F0/4 ... PC 6 = F0/8 on each Access Switch. The notes number the PCs 1 - 18 in both buildings; label yours so you can tell them apart (for example B2-PC1).

Tip: The routers need a serial module for Se0/0/0 (for example a 2911 router with an HWIC-2T module, added while the router is powered off). Use a Serial DCE cable and connect its DCE end (clock icon) to R1.

4. Part 1 - Building 1 configuration

L3-SW1 (HSRP active) - then L3-SW2

Step 1: Hostname and trunks

Configure the links connected to the switch as trunks: F0/1 - F0/3 go to the Access Switches and G0/2 goes to the other L3 switch.

```
enable
configure terminal
hostname L3-SW1
interface range f0/1 - 3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 exit
interface g0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 exit
```

Note: If the switch model rejects the 'trunk encapsulation' line (it supports 802.1Q only), skip that line.

Step 2: Create the VLANs

Create the VLANs and give them names.

```
vlan 10
 name HR
vlan 20
 name Sales
vlan 30
 name Finance
 exit
```

Step 3: SVIs

Each SVI gets its own address. SVIs are shut down by default, so 'no shutdown'.

```
interface vlan 10
 ip address 192.168.10.2 255.255.255.0
 no shutdown
 exit
interface vlan 20
 ip address 192.168.20.2 255.255.255.0
 no shutdown
 exit
interface vlan 30
 ip address 192.168.30.2 255.255.255.0
 no shutdown
 exit
```

Step 4: HSRP (virtual gateways)

One virtual gateway (.1) per VLAN. The higher priority is the active switch. Use 'standby N preempt' with the group number.

```
interface vlan 10
 standby 10 ip 192.168.10.1
 standby 10 priority 110
 standby 10 preempt
 exit
interface vlan 20
 standby 20 ip 192.168.20.1
 standby 20 priority 110
 standby 20 preempt
 exit
interface vlan 30
 standby 30 ip 192.168.30.1
 standby 30 priority 110
 standby 30 preempt
 exit
```

Step 5: Routed port to the router

G0/1 faces R1. 'no switchport' turns it into a Layer 3 routed port.

```
interface g0/1
 no switchport
 ip address 10.10.1.2 255.255.255.252
 no shutdown
 exit
```

Step 6: IP routing and OSPF

Enable routing, then advertise the two router links and the three VLAN networks in OSPF area 0. 'passive-interface' stops OSPF hellos on the user VLANs but the networks are still advertised.

```
ip routing
router ospf 1
 network 10.10.1.0 0.0.0.3 area 0
 network 10.10.1.4 0.0.0.3 area 0
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0
 passive-interface vlan 10
 passive-interface vlan 20
 passive-interface vlan 30
 exit
```

L3-SW2: repeat steps 1 - 6 with these differences

Item	L3-SW1	L3-SW2
Hostname (step 1)	L3-SW1	L3-SW2
SVI addresses (step 3)	192.168.10.2 / 20.2 / 30.2	192.168.10.3 / 20.3 / 30.3
HSRP priority (step 4)	110	100
G0/1 address (step 5)	10.10.1.2 255.255.255.252	10.10.1.6 255.255.255.252

Note: Everything else is identical: the trunks, the VLANs, the virtual gateways (.1) and the OSPF networks. The real SVI address is never .1: .1 belongs only to the HSRP virtual gateway.

Tip: Best practice (not in the notes): make the HSRP active switch the spanning-tree root too: 'spanning-tree vlan 10,20,30 root primary' on L3-SW1 and 'spanning-tree vlan 10,20,30 root secondary' on L3-SW2.

Access-SW1, Access-SW2 and Access-SW3

Access-SW1: trunks, VLANs and floor ports

F0/1 - F0/2 are the trunks to the two L3 switches. The remaining ports (F0/3 - F0/24) are access ports in the floor's VLAN.

```
enable
configure terminal
hostname Access-SW1
interface range f0/1 - 2
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 exit
vlan 10
 name HR
vlan 20
 name Sales
vlan 30
 name Finance
 exit
interface range f0/3 - 24
 switchport mode access
 switchport access vlan 10
 exit
```

Switch	Hostname	Floor ports F0/3 - F0/24
Access-SW1	Access-SW1	switchport access vlan 10 (HR)
Access-SW2	Access-SW2	switchport access vlan 20 (Sales)
Access-SW3	Access-SW3	switchport access vlan 30 (Finance)

Note: Access-SW2 and Access-SW3 use the same commands as Access-SW1. Only the hostname and the access VLAN change.

R1

Step 1: Interfaces

Rename the router and give its interfaces addresses. The serial interface is the DCE side, so it needs the clock rate 64000 (64 kbps).

```
enable
configure terminal
hostname R1
interface g0/1
 ip address 10.10.1.1 255.255.255.252
 no shutdown
 exit
interface g0/0
 ip address 10.10.1.5 255.255.255.252
 no shutdown
 exit
interface s0/0/0
 ip address 10.10.10.1 255.255.255.252
 clock rate 64000
 no shutdown
 exit
```

Step 2: OSPF

```
router ospf 1
 network 10.10.1.0 0.0.0.3 area 0
 network 10.10.1.4 0.0.0.3 area 0
 network 10.10.10.0 0.0.0.3 area 0
 exit
```

5. Part 1 - Building 2 configuration

L3-SW3 (HSRP active) - then L3-SW4

Step 1: Hostname and trunks

```
enable
configure terminal
hostname L3-SW3
interface range f0/1 - 3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 40,50,60
 exit
interface g0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 40,50,60
 exit
```

Note: If the switch model rejects the 'trunk encapsulation' line (it supports 802.1Q only), skip that line.

Step 2: Create the VLANs

```
vlan 40
 name HR
vlan 50
 name Sales
vlan 60
 name Finance
 exit
```

Step 3: SVIs

```
interface vlan 40
 ip address 192.168.40.2 255.255.255.0
 no shutdown
 exit
interface vlan 50
 ip address 192.168.50.2 255.255.255.0
 no shutdown
 exit
interface vlan 60
 ip address 192.168.60.2 255.255.255.0
 no shutdown
 exit
```

Step 4: HSRP (virtual gateways)

```
interface vlan 40
 standby 40 ip 192.168.40.1
 standby 40 priority 210
 standby 40 preempt
 exit
interface vlan 50
 standby 50 ip 192.168.50.1
 standby 50 priority 210
 standby 50 preempt
 exit
interface vlan 60
 standby 60 ip 192.168.60.1
 standby 60 priority 210
 standby 60 preempt
 exit
```

Step 5: Routed port to the router

G0/1 faces R2.

```
interface g0/1
 no switchport
 ip address 10.10.2.2 255.255.255.252
 no shutdown
 exit
```

Step 6: IP routing and OSPF

```
ip routing
router ospf 1
 network 10.10.2.0 0.0.0.3 area 0
 network 10.10.2.4 0.0.0.3 area 0
 network 192.168.40.0 0.0.0.255 area 0
 network 192.168.50.0 0.0.0.255 area 0
 network 192.168.60.0 0.0.0.255 area 0
 passive-interface vlan 40
 passive-interface vlan 50
 passive-interface vlan 60
 exit
```

L3-SW4: repeat steps 1 - 6 with these differences

Item	L3-SW3	L3-SW4
Hostname (step 1)	L3-SW3	L3-SW4
SVI addresses (step 3)	192.168.40.2 / 50.2 / 60.2	192.168.40.3 / 50.3 / 60.3
HSRP priority (step 4)	210	200
G0/1 address (step 5)	10.10.2.2 255.255.255.252	10.10.2.6 255.255.255.252

Note: Everything else is identical. On L3-SW4 also set the allowed list on the G0/2 trunk ('switchport trunk allowed vlan 40,50,60'), like on the other trunks.

Access-SW4, Access-SW5 and Access-SW6

Access-SW4: trunks, VLANs and floor ports

```
enable
configure terminal
hostname Access-SW4
interface range f0/1 - 2
 switchport mode trunk
 switchport trunk allowed vlan 40,50,60
 exit
vlan 40
 name HR
vlan 50
 name Sales
vlan 60
 name Finance
 exit
interface range f0/3 - 24
 switchport mode access
 switchport access vlan 40
 exit
```

Switch	Hostname	Floor ports F0/3 - F0/24
Access-SW4	Access-SW4	switchport access vlan 40 (HR)
Access-SW5	Access-SW5	switchport access vlan 50 (Sales)
Access-SW6	Access-SW6	switchport access vlan 60 (Finance)

R2

Step 1: Interfaces

R2 is the DTE side of the serial link, so no clock rate is needed here.

```
enable
configure terminal
hostname R2
interface g0/1
 ip address 10.10.2.1 255.255.255.252
 no shutdown
 exit
interface g0/0
 ip address 10.10.2.5 255.255.255.252
 no shutdown
 exit
interface s0/0/0
 ip address 10.10.10.2 255.255.255.252
 no shutdown
 exit
```

Step 2: OSPF

```
router ospf 1
 network 10.10.2.0 0.0.0.3 area 0
 network 10.10.2.4 0.0.0.3 area 0
 network 10.10.10.0 0.0.0.3 area 0
 exit
```

Save the configuration

Do this on every router and switch.

```
end
copy running-config startup-config
```

6. Part 2 - Server VLAN, server switch and DHCP relay

Note: DHCP uses broadcasts and broadcasts cannot cross routers. The DHCP server is in Building 1 but many clients are in Building 2, so the L3 switches use DHCP relay ('ip helper-address') to forward the requests to the server. One server can then assign addresses to every department in the enterprise.

Building 1 - L3-SW1 (then L3-SW2)

Steps 1-2: Servers VLAN and trunks

Create VLAN 100 and allow it on the trunk links (F0/1 - F0/3 and G0/2).

```
vlan 100
 name Servers
 exit
interface range f0/1 - 3 , g0/2
 switchport mode trunk
 switchport trunk allowed vlan add 100
 exit
```

Step 3: Gateway and HSRP for the Servers VLAN

```
interface vlan 100
 ip address 192.168.100.2 255.255.255.0
 standby 100 ip 192.168.100.1
 standby 100 priority 210
 standby 100 preempt
 no shutdown
 exit
```

Step 4: Trunk to the server switch (Access-SW7) - L3-SW1 and L3-SW2

F0/4 goes to Access-SW7. It is a trunk that carries only the Servers VLAN. Do this on both L3-SW1 and L3-SW2. The server itself plugs into Access-SW7, not into the L3 switches.

```
interface f0/4
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 100
 exit
end
show interfaces trunk
```

Note: If the switch model rejects the 'trunk encapsulation' line (it supports 802.1Q only), skip that line. F0/24 is not used any more.

Step 5: OSPF - advertise the Servers network

```
configure terminal
router ospf 1
 network 192.168.100.0 0.0.0.255 area 0
 passive-interface vlan 100
 exit
```

Step 6: DHCP relay

Point the three user VLANs to the DHCP server.

```
interface vlan 10
 ip helper-address 192.168.100.10
 exit
interface vlan 20
 ip helper-address 192.168.100.10
 exit
interface vlan 30
 ip helper-address 192.168.100.10
 exit
end
```

Item	L3-SW1	L3-SW2
SVI 100 address (step 3)	192.168.100.2	192.168.100.3
HSRP priority VLAN 100 (step 3)	210	200
Trunk F0/4 to Access-SW7 (step 4)	yes	yes

Tip: Best practice (not in the notes): make the HSRP active switch the spanning-tree root for the Servers VLAN too: 'spanning-tree vlan 100 root primary' on L3-SW1 and 'spanning-tree vlan 100 root secondary' on L3-SW2.

Building 1 - Access-SW1, Access-SW2, Access-SW3

Allow the Servers VLAN on the uplinks

Same commands on all three Access Switches.

```
enable
configure terminal
interface range f0/1 - 2
 switchport mode trunk
 switchport trunk allowed vlan add 100
 exit
```

Building 1 - Access-SW7 (server switch)

Access-SW7: trunks, VLAN and server port

F0/1 and F0/2 are the trunks to L3-SW1 F0/4 and L3-SW2 F0/4. F0/3 is the access port the server connects to. Only the Servers VLAN (100) is needed on this switch.

```
enable
configure terminal
hostname Access-SW7
vlan 100
 name Servers
 exit
interface range f0/1 - 2
 switchport mode trunk
 switchport trunk allowed vlan 100
 exit
interface f0/3
 switchport mode access
 switchport access vlan 100
 exit
end
show vlan brief
show interfaces trunk
```

Port	Connects to	Setting
F0/1	L3-SW1 F0/4	trunk, allowed VLAN 100
F0/2	L3-SW2 F0/4	trunk, allowed VLAN 100
F0/3	Server Fa0	access port, VLAN 100

Note: Access-SW7 is a Layer 2 switch, so it needs no IP address. The 2960 switch supports 802.1Q only, so there is no trunk encapsulation line here.

Note: What is still a single point of failure: Access-SW7 itself, the server's single network card (Packet Tracer servers have only one) and the serial link between R1 and R2. For this lab, the dual-homed server switch is the right level of redundancy.

Building 2 - L3-SW3 (then L3-SW4)

Steps 1-2: Servers VLAN and trunks

```
vlan 100
 name Servers
 exit
interface range f0/1 - 3 , g0/2
 switchport mode trunk
 switchport trunk allowed vlan add 100
 exit
```

Step 3: Gateway and HSRP for the Servers VLAN

```
interface vlan 100
 ip address 192.168.110.2 255.255.255.0
 standby 100 ip 192.168.110.1
 standby 100 priority 210
 standby 100 preempt
 no shutdown
 exit
```

Step 4: OSPF - advertise the Servers network

```
configure terminal
router ospf 1
 network 192.168.110.0 0.0.0.255 area 0
 passive-interface vlan 100
 exit
```

Step 5: DHCP relay

The relay address is still the server in Building 1.

```
interface vlan 40
 ip helper-address 192.168.100.10
 exit
interface vlan 50
 ip helper-address 192.168.100.10
 exit
interface vlan 60
 ip helper-address 192.168.100.10
 exit
end
```

Item	L3-SW3	L3-SW4
SVI 100 address (step 3)	192.168.110.2	192.168.110.3
HSRP priority VLAN 100 (step 3)	210	200

Building 2 - Access-SW4, Access-SW5, Access-SW6

Allow the Servers VLAN on the uplinks

Same commands on all three Access Switches.

```
enable
configure terminal
interface range f0/1 - 2
 switchport mode trunk
 switchport trunk allowed vlan add 100
 exit
```

Save the configuration

Do this on every switch again, including Access-SW7.

```
end
copy running-config startup-config
```

7. Part 2 - Server configuration

Step 1: Static IP settings (Desktop - IP Configuration)

Connect the Server (Fa0) to Access-SW7 F0/3. Then give the server a static address and point its DNS to itself.

Setting	Value
Static IP address	192.168.100.10 (mask 255.255.255.0)
Default gateway	192.168.100.1
DNS server	192.168.100.10 (itself)

Step 2: DHCP service (Services - DHCP)

Turn the service On, fill in each pool and press Add to add the pool to the server. Building 1 pools first, then Building 2 pools.

Pool name	Default gateway	DNS server	Start IP	Mask	Max users
B1-HR	192.168.10.1	192.168.100.10	192.168.10.100	255.255.255.0	100
B1-SALES	192.168.20.1	192.168.100.10	192.168.20.100	255.255.255.0	100
B1-FINANCE	192.168.30.1	192.168.100.10	192.168.30.100	255.255.255.0	100
B2-HR	192.168.40.1	192.168.100.10	192.168.40.100	255.255.255.0	100
B2-SALES	192.168.50.1	192.168.100.10	192.168.50.100	255.255.255.0	100
B2-FINANCE	192.168.60.1	192.168.100.10	192.168.60.100	255.255.255.0	100

Warning: Each pool must be in its own VLAN's subnet. The notes give B2-HR the start IP 192.168.10.100, which is the Building 1 HR subnet, so this guide uses 192.168.40.100 (see the appendix).

Step 3: DNS service (Services - DNS)

Create your own DNS record so users can type a hostname instead of an IP address. Turn the service On, fill in the record and press Add.

Field	Value
Name	company.local
Address	192.168.100.10

Step 4: HTTP service (Services - HTTP)

Enable the HTTP service to host the internal website (the company home page). Users open http://company.local in a browser.

Note: The DHCP pools hand out 192.168.100.10 as the DNS server, so every PC can resolve company.local.

8. End devices

Device	VLAN	IP address	Gateway	DNS
Building 1 Printer 1 (HR)	10	192.168.10.20 /24 (static)	192.168.10.1	192.168.100.10
Building 1 Printer 2 (Sales)	20	192.168.20.20 /24 (static)	192.168.20.1	192.168.100.10
Building 1 Printer 3 (Finance)	30	192.168.30.20 /24 (static)	192.168.30.1	192.168.100.10
Building 2 Printer 1 (HR)	40	192.168.40.20 /24 (static)	192.168.40.1	192.168.100.10
Building 2 Printer 2 (Sales)	50	192.168.50.20 /24 (static)	192.168.50.1	192.168.100.10
Building 2 Printer 3 (Finance)	60	192.168.60.20 /24 (static)	192.168.60.1	192.168.100.10
All 36 PCs	10 - 60	DHCP (automatic, .100 and up)	HSRP address (.1)	192.168.100.10

Note: Printers get static addresses (.20), outside the DHCP range (.100 and up), so everyone finds them at the same address. Computers use DHCP from the central server.

9. Quick verification

Check	Command / test
VLANs and port assignment	show vlan brief (all switches)
Trunks up, VLANs allowed	show interfaces trunk (all switches)
Server switch	On Access-SW7: show interfaces trunk (F0/1 and F0/2 trunking VLAN 100) and show vlan brief (F0/3 in VLAN 100)
HSRP active / standby	show standby brief (L3 switches: one Active and one Standby per VLAN)
Routed links up	show ip interface brief (routers and L3 switches)
OSPF neighbors	show ip ospf neighbor (routers and L3 switches)
Routes to the other building	show ip route (R1 sees 192.168.40.0 - 60.0 as OSPF routes; R2 sees 192.168.10.0 - 30.0)
DHCP relay set	show ip interface vlan 40 (shows 'Helper address is 192.168.100.10')
PCs get an address	On each floor: IP configuration - DHCP gives 192.168.x.100 or higher (not 169.254.x.x)
Gateway and cross-building	Ping the gateway (.1); ping a PC in the other building
DNS and web	From a PC browser open http://company.local (the home page loads)
Server redundancy	Run the failover test below

Server failover test

This proves the server survives the loss of one multilayer switch.

Step	What to do	Expected result
1	Run show standby brief on L3-SW1 and L3-SW2	L3-SW1 is Active and L3-SW2 is Standby for VLAN 100
2	Power off L3-SW1 in Packet Tracer	The links to L3-SW1 go down
3	Run show standby brief on L3-SW2	L3-SW2 is now Active for VLAN 100
4	From a PC in each building: ping 192.168.100.10 and renew DHCP	Ping replies and the PC gets an address
5	Power L3-SW1 back on	L3-SW1 becomes Active again (preempt)

Note: Give the network a short time after each change: spanning tree and HSRP need a moment before the results are stable.

10. Troubleshooting: HR PCs get no IP address

A person from HR called and said none of the computers could connect to the network.

Step 1: Verify the problem

Open the IP configuration of one of the HR PCs and request an IP address with DHCP. The PC gets a 169.254.x.x address (for example 169.254.94.238). This is an APIPA address: when a PC cannot reach a DHCP server, it gives itself one of these addresses.

Step 2: Check who is affected

Repeat step 1 on one PC on each floor (HR, Sales, Finance). The Sales and Finance PCs get addresses by DHCP normally, so only the HR floor (Building 2, VLAN 40) is affected.

Step 3: Check the DHCP configuration on the server

Open Services - DHCP on the server and check the pools. The B2-HR pool must exist, with gateway 192.168.40.1 and start IP 192.168.40.100.

Step 4: Check Access-SW4 (the switch HR is connected to)

```
enable
show vlan brief
show interface trunk
```

Steps 5-6: Check L3-SW3 and L3-SW4

Run the same two commands on each of them.

```
enable
show interface trunk
show ip interface brief
```

Where	Command	Look for
Access-SW4	show vlan brief	The HR PCs are in VLAN 40
Access-SW4	show interface trunk	VLAN 40 is allowed on the trunk
L3-SW3 / L3-SW4	show interface trunk	VLAN 40 is allowed on the trunk
L3-SW3 / L3-SW4	show ip interface brief	The VLAN 40 gateway (Vlan40) is up

Step 7: Check the running-config on L3-SW3

Look at the VLAN 40 interface.

```
enable
show running-config
```

Problem found	Why it matters
The 'ip helper-address' command is not enabled on VLAN 40 of L3-SW3	DHCP requests are broadcasts. Without the helper address they cannot travel from Building 2 to the DHCP server in Building 1, so the HR PCs in Building 2 cannot get an IP address.

Step 8: Fix VLAN 40 on L3-SW3

Re-configure the VLAN 40 interface with the relay address, then check it.

```
enable
configure terminal
interface vlan 40
 ip helper-address 192.168.100.10
end
show running-config interface vlan 40
```

Step 9: Verify the fix

Go onto the PCs on the HR floor, open the IP configuration and request an IP address by DHCP. They now get 192.168.40.x addresses.

11. Troubleshooting: duplicate address on a VLAN

Example: L3-SW2 keeps printing the message below and it does not stop.

```
%IP-4-DUPADDR: Duplicate address 192.168.30.1 on Vlan30, sourced by 0000.0C07.AC1E
```

Clue	Meaning
MAC 0000.0C07.ACxx	This is the HSRP virtual MAC. The last two hex digits are the group number: 1E hex = 30, so it is group 30 (VLAN 30).
Cause	A switch has the HSRP virtual IP (.1) as the real address of its own SVI. It clashes with the virtual gateway. The real SVI must be .2 (L3-SW1) or .3 (L3-SW2).

Step 1: Check both L3 switches

```
show ip interface brief
show running-config interface vlan 30
```

Switch	Vlan30 real IP	Virtual IP
L3-SW1	192.168.30.2	standby 30 ip 192.168.30.1
L3-SW2	192.168.30.3	standby 30 ip 192.168.30.1

Step 2: Fix the SVI (example: L3-SW2, VLAN 30)

```
enable
configure terminal
interface vlan 30
 ip address 192.168.30.3 255.255.255.0
 standby 30 ip 192.168.30.1
 standby 30 priority 100
 standby 30 preempt
 no shutdown
 end
```

Check VLANs 10 and 20 on L3-SW2 in the same way (.3 each, and 192.168.100.3 for VLAN 100). On Building 2 the same rule applies to L3-SW4 (.3).

Step 3: Refresh the interface and clear the ARP table

```
configure terminal
interface vlan 30
 shutdown
 no shutdown
 end
clear arp-cache
```

Step 4: Check the Finance floor devices

No device on the floor may have a static 192.168.30.1. Printer 3 must be 192.168.30.20 and the PCs must use DHCP.

Step 5: Verify

Press Enter and wait. No new DUPADDR lines should appear. Then run show standby brief: one Active and one Standby per VLAN, virtual IP .1.

> **Tip:** To stop log messages from interrupting what you type (this hides the interruption only, it does not fix the cause): 'line console 0' then 'logging synchronous'.

Appendix: corrections made to the original notes

In the notes	Used in this guide
'switchport allowed vlan 10,20,30' (trunks)	switchport trunk allowed vlan 10,20,30
'HRSP'	HSRP
'standby preempt' (no group number = group 0 only)	standby 10 preempt (one per group: 10, 20, 30 ...)
Router mask '255.255.2555.252' (R1 and R2)	255.255.255.252
'clock rate 64KPS'	clock rate 64000 (64 kbps), on R1 only (DCE)
OSPF 'network 192.168.100 ...' and '192.168.110 ...'	network 192.168.100.0 and 192.168.110.0
B2-HR DHCP pool start IP '192.168.10.100'	192.168.40.100 (VLAN 40 subnet)
Building 2 cabling copied from Building 1 (Access-SW1 - 3, L3-SW5, L3-SW6)	Access-SW4 - 6 and L3-SW4
Printer 2 and Printer 3 'to Access-SW1 F0/9'	Access-SW2 F0/9 and Access-SW3 F0/9 (both buildings)
L3-SW4 G0/2 trunk had no allowed VLAN list	Added 'switchport trunk allowed vlan 40,50,60'
Part 2 L3-SW2 / L3-SW4 steps numbered 1, 2, 3, 6, 7	Steps renumbered
'##' comments after the troubleshooting commands	Moved to the 'Look for' table
'end' then 'do show run' (do only works in config mode)	show running-config interface vlan 40
'ip helper-address isn't enabled on VLAN' (number missing)	VLAN 40
Server connected only to L3-SW1 F0/24 (single link)	Server on Access-SW7 F0/3, dual-homed to L3-SW1 F0/4 and L3-SW2 F0/4

Added so the commands work (not written in the notes): the serial module and DCE cable tip, 'copy running-config startup-config', the L3-SW2 / L3-SW4 and Access Switch difference tables, the spanning-tree root tip, the server subnet mask (255.255.255.0), and the verification commands.

Added after feedback on the design (not in the notes): Access-SW7 and its cabling and configuration, the F0/4 trunks on L3-SW1 and L3-SW2, the spanning-tree root tip for VLAN 100, the server failover test, and section 11 (duplicate address troubleshooting).